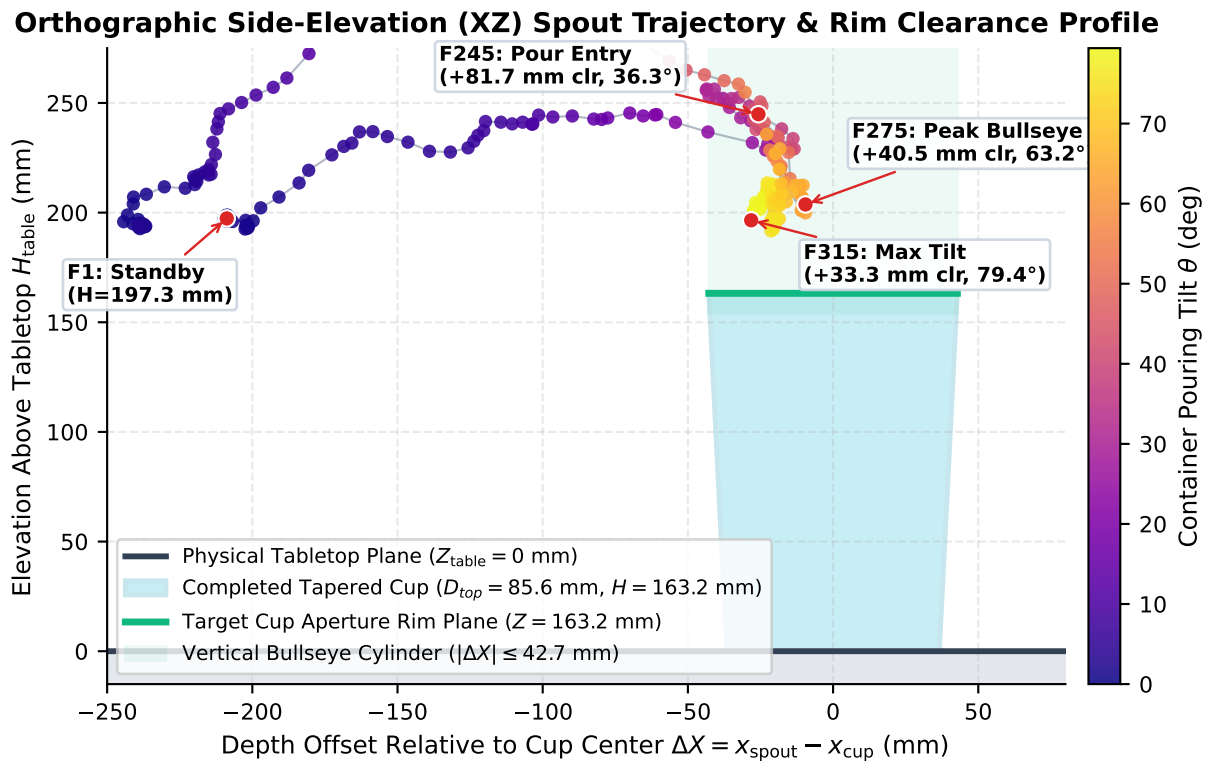

Orthographic Side-Elevation (XZ) Spout Trajectory & Rim Clearance Profile
F245: Pour Entry
(+81.7 mm clr, 36.3°)
F275: Peak Bullseye
(+40.5 mm clr, 63.2°
F315: Max Tilt
(+33.3 mm clr, 79.4°)
F1: Standby
(H=197.3 mm)
Physical Tabletop Plane ($Z_{table} = 0$ mm)
Completed Tapered Cup ($D_{top} = 85.6$ mm, $H = 163.2$ mm)
Target Cup Aperture Rim Plane ($Z = 163.2$ mm)
Vertical Bullseye Cylinder ($|\Delta X| \leq 42.7$ mm)
Elevation Above Tabletop H_{table} (mm)
Depth Offset Relative to Cup Center $\Delta X = x_{spout} - x_{cup}$ (mm)
Container Pouring Tilt θ (deg)
250
200
150
100
50
0
−250
−200
−150
−100
−50
0
50
70
60
50
40
30
20
10
0